

The Privacy Problem Nobody's Blocking

Most privacy tools protect the wrong layer

Trackers, VPNs, ad-IDs, cookies — all of this is about **collection**. Stop the data from leaving your device, and you're "protected." That's the entire premise behind DNS filters, VPN-slot blockers, incognito mode.

But collection was never really the threat. **Inference** is.

The paradox: privacy tools are themselves a fingerprint

Here's the part almost nobody talks about. The moment you install a tracker blocker, use a VPN, or run uBlock Origin, you don't disappear — you become a *rarer* pattern. Fewer than 5% of users block trackers. Fewer still run a VPN alongside it. The combination of "blocks trackers + runs specific DNS + specific browser fingerprint size + specific screen resolution + specific installed-font list" is often **more unique**, not less, than doing nothing at all.

This is called **anonymity set collapse**: privacy tools only work when enough other people use the *same* tool, in the *same* configuration, at the *same* time. A VPN with 50,000 users hides you in a crowd. A self-hosted, uniquely-configured Pi-hole with a custom blocklist you wrote yourself does the opposite — it makes you the only person on earth with that exact signature.

Tor is the one tool built around this insight deliberately: everyone's browser is *forced* to look identical (same window size, same user-agent, same everything) specifically so the anonymity set stays large. Almost no other privacy tool does this. Most "privacy" software actually optimizes for *individual protection* while accidentally *destroying the crowd* that protection depends on.

Correlation attacks don't need your data — just your shape

Even with zero trackers, zero cookies, zero identifiers: if your phone connects to the cell tower at 8:03am from location A and 5:47pm from location B, five days a week, for a year — that's a full behavioral profile, built from metadata your carrier is required to have to route calls. No tracker installed. No consent dialog. No blocklist can touch it, because it's not "tracking" in the legal or technical sense — it's **inference from unavoidable operational data**.

This is the real ceiling on personal privacy tooling: you can block *collection*, but the *shape* of your existence (timing, cadence, correlation across otherwise-anonymous signals) leaks information no blocklist addresses, because it was never collected by an app — it's a byproduct of the network existing at all.

What this means practically

- Building a personal anti-tracker (like the one we discussed) genuinely helps against **collection** — real, worthwhile, still block it.
- But uniqueness of your setup is itself a cost. A blocklist *you* alone maintain is more identifying than a popular shared one.
- True protection against inference isn't a tool — it's **joining a crowd that already exists** (Tor, a widely-used VPN, a mainstream blocklist) rather than engineering a bespoke one-person setup.
- The strongest privacy move available to most people isn't a new app. It's making your device *boringly identical* to a million other devices.

Privacy, in other words, isn't something you build alone. It's something you can only borrow from everyone doing the same thing you are.

Free & Open: A Real Ad + Tracker Blocker for Android

The offer

A system-wide ad, tracker, phishing, and malware blocker for Android — **100% free**, no subscription, no lifetime license fee, no premium tier locked behind a paywall. Built entirely on open blocklists and free Android APIs, distributed outside the Play Store to avoid the policy restrictions that limit what paid tools like AdGuard can do on their free tier.

Target: **85-95% blocking effectiveness**, in line with paid competitors — for €0.

What it blocks

Layer	Source	Blocks
Ads & trackers	StevenBlack hosts list + EasyList + EasyPrivacy	Ad networks, analytics SDKs, tracking pixels
Malware & phishing	UT1 Toulouse Capitole blacklist (ads , phishing , cryptojacking ,	Known bad domains, crypto-mining scripts,

Layer	Source	Blocks
	marketingware categories)	phishing pages
Stalkerware	UT1 stalkerware category	Covert monitoring apps calling home
First-party ad slots (partial)	Cosmetic filtering via in-app browser	Ads served from the same domain as page content

How it works (no cost at any layer)

1. **VpnService local filter** — Android's built-in API lets an app act as a local traffic filter without needing a real VPN server. Free, no infrastructure to pay for or maintain.
2. **Merged blocklist, compiled for mobile** — the raw lists (millions of domain entries combined) are deduplicated and compiled into a compact Bloom filter, so lookups are instant and the whole thing fits in a few MB of storage instead of hundreds.
3. **Per-app stats dashboard** — see exactly what's being blocked and where, so you can judge the real scale of tracking on your own phone (not just take a vendor's word for it).

4. **Free distribution** — sideloaded APK or F-Droid, not the Play Store, so it isn't limited by Google's policy restrictions on ad-blocking VPN apps.

What you don't get, and why (honesty over hype)

- **Not 100% blocking.** First-party tracking (an app logging your own usage of *itself*) and device fingerprinting aren't stoppable by any blocklist — free or paid. Anyone promising 100% is overselling.
- **No HTTPS/SNI deep inspection at launch.** That requires installing a root certificate on the device — powerful, but a bigger trust ask. Left as an optional advanced feature, not a default.
- **No dedicated support team or SLA.** This is a personal/open project, not a company. Updates depend on list maintenance being kept current — a real, ongoing task, not a one-time build.

Additional features worth building in

Beyond core ad/tracker blocking, a few features would round this out into something closer to a full protection suite — still free, all achievable with the same VpnService foundation:

- **Per-app firewall** — allow/deny internet access per app entirely, not just tracker domains. Useful for apps that don't need network access at all (offline games,

utilities) or ones you want fully cut off (e.g. Disney+ sitting unused but generating background traffic). This reuses the same VpnService hook already doing the blocking, so it's a UI/rules layer more than new infrastructure.

- **Per-app data usage breakdown** — how much traffic each app sends, not just how many tracker attempts were blocked. Surfaces apps quietly burning data/battery in the background even after tracking is blocked.
- **Scheduled/profile-based rules** — e.g. a stricter profile for public WiFi, a relaxed one at home; or a "work mode" that only allows a whitelisted set of apps online.
- **DNS-level malware/phishing warnings** — real-time alert (not just silent block) when an app tries to reach a known-bad domain, so you actually see the moment it happens rather than only checking a weekly total.
- **Export/backup of block logs** — since you're already manually tracking 7-day totals from DDG, built-in CSV/JSON export would remove that manual step entirely.
- **Category toggles from the UT1 list** — since UT1 covers more than ads (gambling, adult content, social media, etc.), exposing these as optional user-toggable categories turns the same engine into a parental-control tool for free, without extra blocklist work.

OPNsense-grade filtering, phone-grade simplicity

Tools like OPNsense/pfSense with SquidGuard (where the UT1 blacklist comes from) give serious filtering power — but they require a dedicated router or box, real networking knowledge, and ongoing maintenance most people will never take on. That's simply out of reach for a normal phone user who just wants their apps to stop leaking data.

This project brings the same underlying idea — intercept traffic, check it against a blacklist, drop the bad stuff — down to a single Android app. No router to flash, no VLANs to configure, no command line. Install, tap enable, done. The filtering engine is the same category of technology; the barrier to actually using it is what disappears.

DDoS & intrusion protection (IPS)

Blocking known-bad domains handles outbound tracking and malware, but it doesn't address attacks aimed *at* the device — malicious inbound connection floods, port scanning, or exploit attempts targeting apps listening on the network (common on public WiFi). Adding a lightweight IPS/DDoS layer closes that gap, and reuses [[realguard]]'s design as the blueprint:

- **Connection-rate anomaly detection** — flag and throttle abnormal bursts of inbound connection attempts, the mobile equivalent of a DDoS mitigation rule on a router

- **Port scan detection** — recognize sequential/rapid probing patterns and block the source before it maps open services
- **Signature-based intrusion detection** — lightweight pattern matching against known exploit/attack signatures (a mobile-scale Snort/Suricata concept, not the full engine)
- **Public WiFi auto-hardening** — automatically tighten IPS sensitivity when the device joins an untrusted/open network, relaxing it back on trusted home WiFi
- **Alert log, not just silent block** — same principle as the phishing alerts above: the user sees *that* an intrusion attempt happened, not just a number in a weekly total

This turns the app from "ad/tracker blocker" into a genuine consumer-grade security layer — outbound protection (blocklists) plus inbound protection (IPS/DDoS), the two halves that a router-based OPNsense setup would normally split across firewall rules and IDS/IPS packages, now combined in one free app.

What actually makes it one of a kind

Firewall, IPS, DDoS protection, malicious-app scanning — all of that is table stakes now; AdGuard and others already do versions of it. The features below are the ones nobody in this space is shipping, because they require either a local reasoning layer or a philosophy most privacy tools don't have:

- **Inference-risk warning (from the anonymity-set problem above)** — the app actively calculates how unique your specific combination of blocked trackers, DNS setup, and device signals makes you, and warns you when your "protection" has started making you *more* identifiable, not less. No blocker on the market does this — they all optimize for block count and ignore that the config itself is a fingerprint.
- **Local AI correlation and plain-language explanations** — not a copy of [[anora]], but a small AI built from scratch specifically for this app: a narrow, purpose-trained on-device model (or lightweight classifier + template-based explanation engine, whichever proves lighter for a phone) that only ever reasons about tracking/blocking data. No dependency on the main Anora install, no general-purpose capability to maintain, no cloud call — just enough intelligence to turn raw block logs into insight: *"Critical Strike has contacted 11 ad/analytics companies in the last hour — this is unusually high even for a free game."* Being purpose-built (not a trimmed general assistant) keeps it small enough to ship inside the app itself and makes its behavior fully auditable — you can point at exactly what it was trained/built to do, which is itself part of the "provably zero telemetry" story below.
- **Supply-chain / behavior-change detection** — flag when an already-installed app suddenly starts contacting new domains after an update (a common sign of a compromised SDK or a company quietly

adding trackers). Nothing consumer-facing currently watches for *change* in an app's behavior over time, only static blocklists.

- **"Who and why" transparency layer** — instead of just showing a blocked domain, show which company owns it and what category of data they typically collect (ad ID, location, device fingerprint, etc.), sourced from open trackerdb-style data. Turns the tool into an education layer, not just a silent filter.
- **Zero telemetry, provably** — many "free" tools still phone home their own usage analytics. This one publishes its own network traffic as open source specifically so users (or you, as the developer) can verify nothing leaves the device — turning the "free and open" claim from a promise into something checkable.

These five are the actual differentiators — everything else in this doc (firewall, IPS, ad/tracker blocking) is good, necessary, and free, but not unique. This section is.

The most important requirement: built for non-technical users

Every feature above is worthless if it needs a technical person to configure it. OPNsense, Pi-hole, even AdGuard's advanced options all lose normal users at the setup screen — that's the entire reason this project exists instead of "just tell people to use OPNsense." So the design priority above every feature is: **one tap, zero configuration required, plain language only.**

- **Single on/off switch by default.** Firewall, IPS, DDoS protection, blocklists — all enabled together the moment the app is installed. No categories to pick, no rules to write, no ports to understand.
- **Plain-language explanations, never jargon.** No "SNI," "Bloom filter," "IPS," or "DNS" in the user-facing UI — that's implementation detail for the doc, not the app. The AI explanation layer exists precisely so the user reads *"Critical Strike is contacting 11 companies you don't know"* instead of a log line.
- **Advanced settings exist, but are hidden by default.** Power-user controls (category toggles, firewall per-app rules, IPS sensitivity) live one tap deeper for people who want them — never required to get full protection out of the box.
- **No setup wizard.** Install, tap enable, done — the entire onboarding is the two things a non-technical person will actually do.
- **Notifications explain impact, not mechanics.** "Blocked 40 tracking attempts today" beats "40 DNS queries dropped." The number matters to a normal user; the mechanism doesn't.

This is the real differentiator that ties every other feature together: the protection is OPNsense-grade, but the experience has to feel like flipping a single light switch — because that's the only way it actually reaches the people who need it and can't set up a router themselves.

One app, not three products stitched together

The old model — separately configuring a proxy (squid), a DDoS mitigation layer, and an IPS as three distinct systems with their own setup steps — is exactly what this project avoids, on the build side as much as the user side. Blocklists, firewall, and IPS/DDoS all run on the **same VpnService interception point**; they're different rule layers checked against the same traffic stream, not three separate services a developer has to wire together and keep in sync.

Practically: one app, one traffic pipeline, one settings model. Blocklist matching, per-app firewall rules, and connection-anomaly detection for IPS/DDoS all just add on to the same "check this connection, allow or drop it" decision — not three engines with three configs to maintain. That's what keeps both the build and the user experience genuinely simple instead of a squid+IPS+DDoS setup wearing an app icon.

Why free is realistic here, not just a promise

Every paid competitor (AdGuard, Total Adblock, Surfshark CleanWeb) is built on the *same* free, community-maintained blocklists underneath their pricing. The lists are the actual engine — the subscription pays for polish, infrastructure, and support, not for better raw blocking

power. Skipping the subscription costs you convenience, not effectiveness.

Comparison snapshot

	This project	AdGuard Free	AdGuard Premium
Cost	€0	€0	€2.49-5.49/mo or €79.99 lifetime
System-wide (non-browser apps)	✓	✓	✓
Cosmetic/first-party filtering	Partial (in-app browser)	Browser only	Full, HTTPS inspection
Stalkerware/phishing blocking	✓ (via UT1)	✓	✓
Anonymity-set awareness	✓ (unique feature)	✗	✗
Per-app firewall	✓	Premium only	✓
IPS / DDoS / intrusion protection	✓	✗	✗

	This project	AdGuard Free	AdGuard Premium
Play Store install	✗ (sideload/F-Droid)	Limited version	✗ (sideload)

Why free, on purpose

Every tool in this space — AdGuard, Total Adblock, Surfshark CleanWeb — charges for something built on the same open, community-maintained blocklists anyone can download for free. The blocking power isn't the product being sold; convenience and polish are. That's a legitimate business, but it also means privacy protection ends up gated behind a subscription for people who can't or won't pay — often the same people with the least power to push back on being tracked in the first place.

This project exists to close that gap: give away the actual protection, not a watered-down free tier designed to nudge people toward upgrading. No ads to fund it, no data sale, no "free forever, except the features that matter." If it's not free and effective for everyone, it's not really the point.

Bottom line

A genuinely free, effective (85-95%) blocker is achievable by standing on the same open lists the paid tools already

use — the gap between free and paid isn't blocking power, it's convenience and polish. Built right, this doubles as both a real privacy tool and a strong portfolio piece — one that gives back rather than gatekeeps.